Step 1: Problem Analysis & System Diagram

Problem Statement
To address the needs of animal shelters, we need to develop a cost-effective pet feeder system that can automatically provide food for cats and dogs. The system should be capable of dispensing food at scheduled times, ensuring that animals are fed consistently and on time. Additionally, it must incorporate weight sensors to monitor food consumption, allowing for better tracking of each animal's eating habits.

This feeder system should also be equipped with an alert mechanism to notify shelter staff if there are any issues. For instance, if no food is consumed by the animals or if there is an insufficient supply of food, the system should automatically send a notification to inform the staff, enabling them to take necessary actions promptly. This will ensure that the animals' nutritional needs are met and any disruptions in feeding are quickly addressed.

In conclusion, a reliable and automated feeding system would greatly improve the efficiency of feeding routines in animal shelters while minimizing human intervention. This approach will not only save time and labor but also ensure the well-being of the animals by providing them with regular meals and timely monitoring.

Assumptions & Limitations:

- The feeder uses only one type of dry pet food.
- Feeding times are fixed but can be programmed by the user.
- The system can detect if food remains in the bowl after feeding. If the bowl weight changes by less than 5 g after 10 minutes, an “uneaten” alert is triggered.
- Limited memory is available for feeding schedules. It can store up to 7 feeding times and does not keep historical data.
- Runs on 12V DC power with no battery backup and cannot send alerts on power failure.
- Requires manual refilling when the food container is empty.

Inputs & Outputs:
Inputs: Feeding times, food-level sensor (binary), bowl weight (grams).
Outputs: Servo motor (dispense), buzzer/LED (alerts), LCD display (status and notifications).

Block Diagram of the Pet-Feeder-System

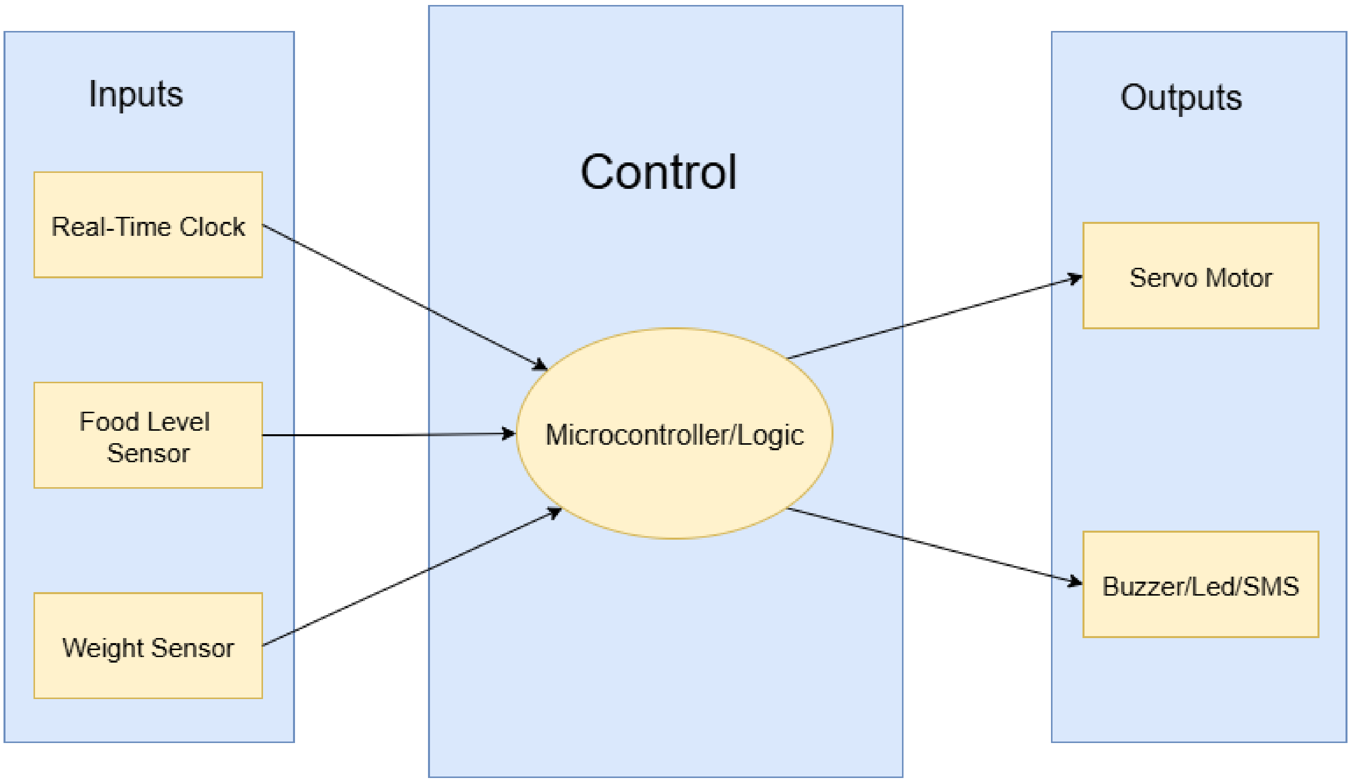

Figure 1: System Diagram